

Faculty Vitae: Belinda S Akpa

1. Education

- B.A. Chemical Engineering, University of Cambridge (UK), 2002
- M.Eng. Chemical Engineering, University of Cambridge (UK), 2002
- Ph.D. Chemical Engineering with Advanced Chemical Engineering Practice, University of Cambridge – Cambridge/MIT Institute, 2007

2. Academic Experience

- Department of Chemical Engineering, University of Illinois-Chicago
Professor, 2026-present
- Department of Chemical & Biomolecular Engineering, University of Tennessee-Knoxville
Associate Professor, 2020-2026
- Department of Molecular Biomedical Sciences, North Carolina State University
Assistant Professor, 2015-2020
- Department of Chemical Engineering, University of Illinois-Chicago
Assistant Professor, 2009-2015
- Department of Chemical Engineering, University of Illinois-Chicago
Post Doctoral Research Fellow, August 2007-August 2009

3. Non-Academic Experience

- Oak Ridge National Laboratory, Biosciences Division
Senior Staff Scientist

4. Certifications or professional registrations

- Not applicable

5. Current membership in professional organizations

- American Institute of Chemical Engineers 1999-present

6. Honors and awards

- Chemical and Biomolecular Engineering Teaching Award, UTK 2025
- Graduate Student Senate Award for Outstanding Graduate Research Mentor, UTK 2024 (student nominated)
- American Institute of Chemical Engineering, MAC Distinguished Service Award, 2019
- Harold Simon Award for Excellence in Teaching, 2015, College of Engineering, UIC (highest teaching distinction in the College of Engineering)
- UIC College of Engineering Undergraduate Advising Award, 2012
- UIC College of Engineering Teaching Award, 2012 & 2013

7. Service activities

- Leadership team, Computational core, Humanity Unlocking Biomaterials (HUB) NIBIB U24 Biomaterials Coordinating Center (2024-present)
- Steering Committee for NSF Research Coordination Network: Finding Your Inner Modeler (2020-present)
- Expert Mentor, National Cancer Institute S³ (synthetic, systems, and spatial) Innovation Lab (June 2025)

Organizing Committee, Institute for Computational and Experimental Research in Mathematics workshop “Fostering Cross-Disciplinary Collaboration in Biology, Medicine, and Computational Science” (2025)
American Institute of Chemical Engineers, Career & Education Operating Council (2020-2024)
Member, Department of Energy Office of Science Pandemic Response Roundtable (2022)
American Institute of Chemical Engineers, Societal Impact Operating Council (2018-2020)

8a. Selected Publications

5. Zheng S., Chiang Y-H., Aldarondo D., Fox Z.R., Akpa B.S.[†], and Wayne E.[†] Capturing the dynamics of STAT6 macrophage polarization using bioluminescence temporal spectrometry. *Cell Systems*. <https://doi.org/10.1016/j.cels.2026.101643> (2026)
4. Farahat M., Flaherty D.T., Fox Z.R., Akpa B.S.[†] Prediction variability in physiologically based pharmacokinetic modeling of tissue disposition under deep uncertainty. *npj Systems Biology and Applications*. <https://doi.org/10.1038/s41540-026-00739-7> (2026)
3. Hodgens C.D., Flaherty D.T., Pullen A-M., Khan I., English N.J., Gillan L.M., Rojas-Pierce M., Akpa B.S.[†] Model-based inference of a dual role for HOPS in regulating guard cell vacuole fusion. in *Silico Plants*. pp. diae015 (2024)
2. Blanchard A.E.[†], Bhowmik D., Fox Z.R., Gounley J., Glaser J., Akpa B.S., Irle S. Adaptive Language Model Training for Molecular Design. *Journal of Cheminformatics*. 15(1):59 (2023)
1. Hodgens C., Akpa B.S.[†], Long, T.A.[†] Solving the puzzle of Fe homeostasis by integrating molecular, mathematical, and societal models. *Current Opinion in Plant Biology*. 64:102149 (2021)

8b. Selected Presentations

5. Keynote. Boston University Bioinformatics Student-Organized Symposium. Jun 2026. “Model-driven discovery uncovers quantitative answers in the qualitative data describing plant Fe homeostasis”
4. Plenary. Computational Modeling in Biology Network (COMBINE) Workshop. Oct 2025. “Systems modeling integrates scarce, discordant observations to uncover quantitative answers in the qualitative data describing plant Fe homeostasis”
3. Keynote. Data Analysis and Deep Learning Track. Biomedical Engineering Society Annual Meeting. Oct 2023. “Enabling inverse design of molecules to drive defined physiological outcomes”
2. Humanity Unlocking Biomaterials Coordinating Center Annual Meeting. April 2025. “Where should we begin? Building a relationship with modeling and computational thinking.”
1. Interagency Modeling & Analysis/Multiscale modeling Consortium (IMAG/MSM) working group on multiscale modelling & viral pandemics. March 2024. “Every model has its place: combining systems, data-driven, and generative models for therapeutic design”